

# Appendix A - Abrolhos Shallow Bank Marine Park Natural Values Report

North-west Marine Parks Network

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## Natural Values

Understanding of natural values

## Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

| Date         | Campaign name                                   | Method | Areas    | Samples | Habitat | Fish  |        |
|--------------|-------------------------------------------------|--------|----------|---------|---------|-------|--------|
|              |                                                 |        |          |         | Count   | Count | Length |
| May 2021     | 2021-05 Abrolhos<br>BOSS                        | BOSS   | NPZ, SPZ | 46      | 46      | -     | -      |
| May 2021     | 2021-05 Abrolhos<br>stereo-BRUVs                | BRUV   | NPZ, SPZ | 24      | 24      | 23    | 23     |
| Feb-Mar 2025 | 2025-02<br>Yamatji-Shallow-Bank<br>stereo-BRUVs | BRUV   | NPZ, SPZ | 120     | 115     | 114   | 109    |

## Benthic ecosystem extent

### Observations

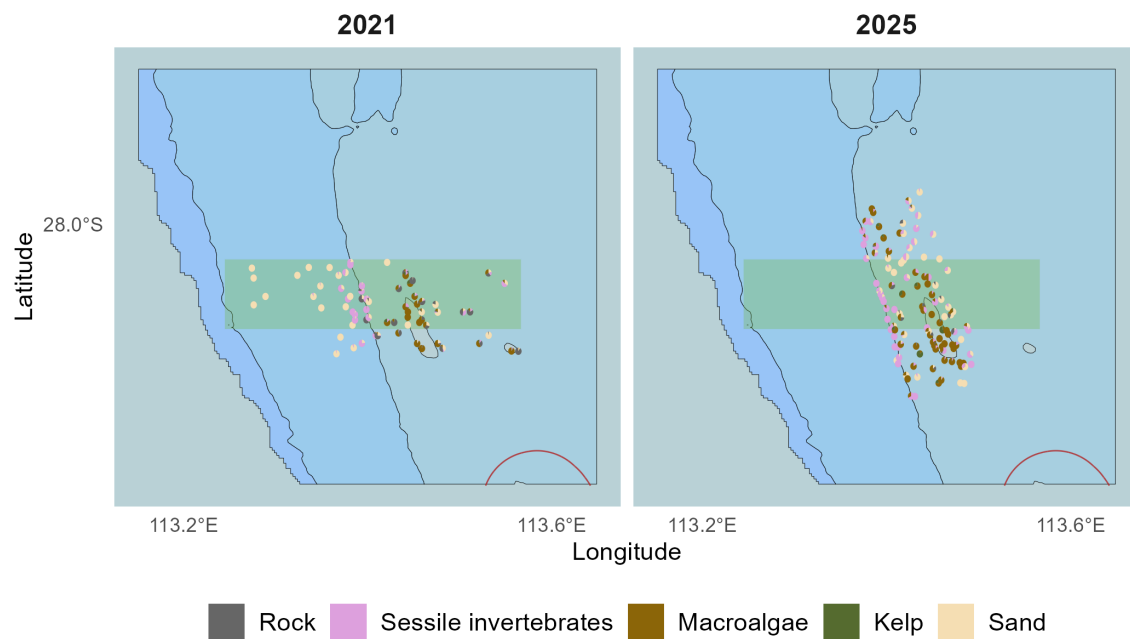


Figure 1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts, pooled across the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Dominant habitat types

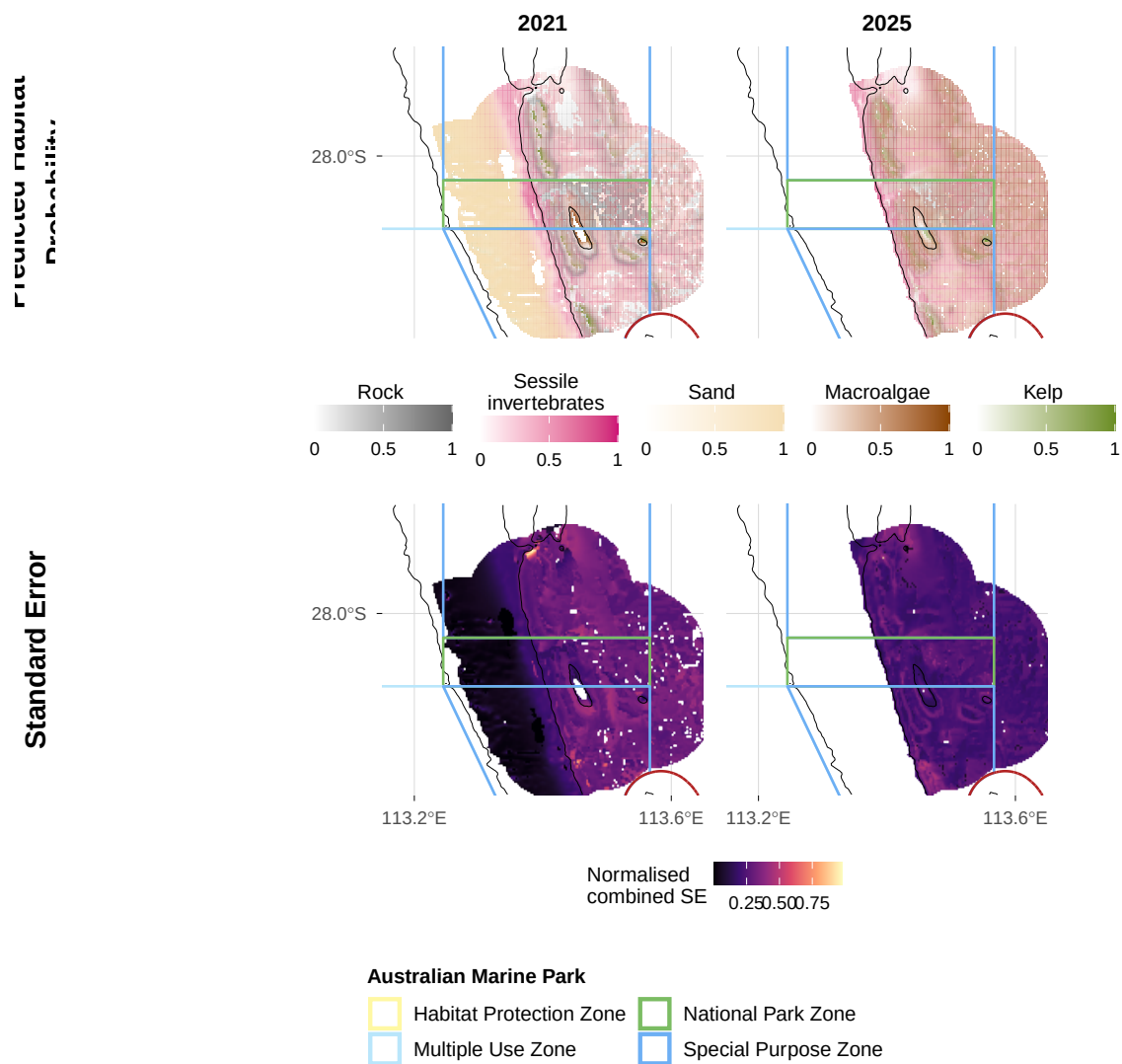


Figure 2: Predicted dominant habitat type from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Habitat distribution and probability

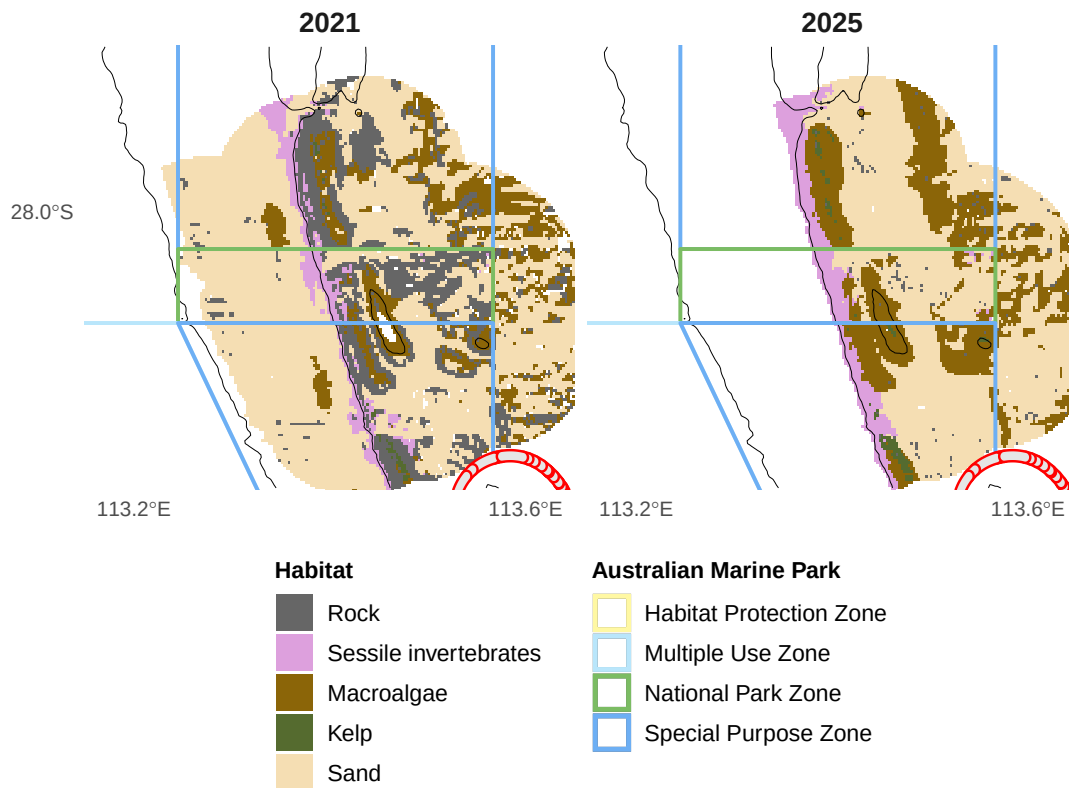


Figure 3: Predicted habitat class distribution from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Functional reef distribution and probability

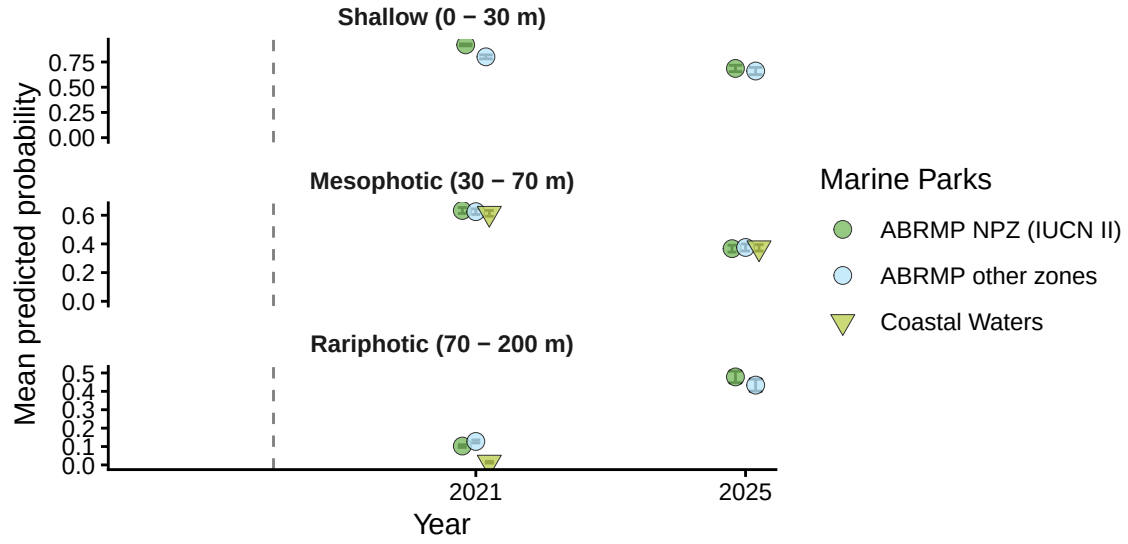


Figure 4: Functional reef: benchmark plots by zone and depth contour from stereo-BRUV and drop-camera groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

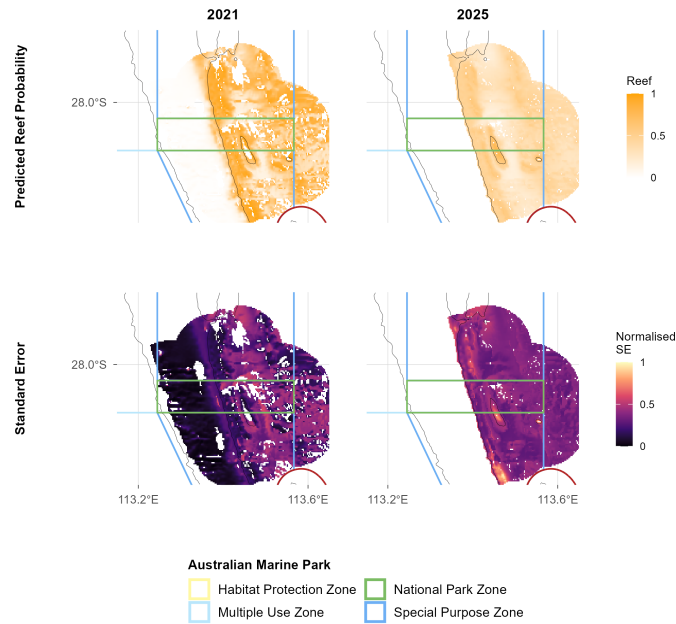


Figure 5: Predicted reef distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Benchmarks of benthic natural values

### Sand

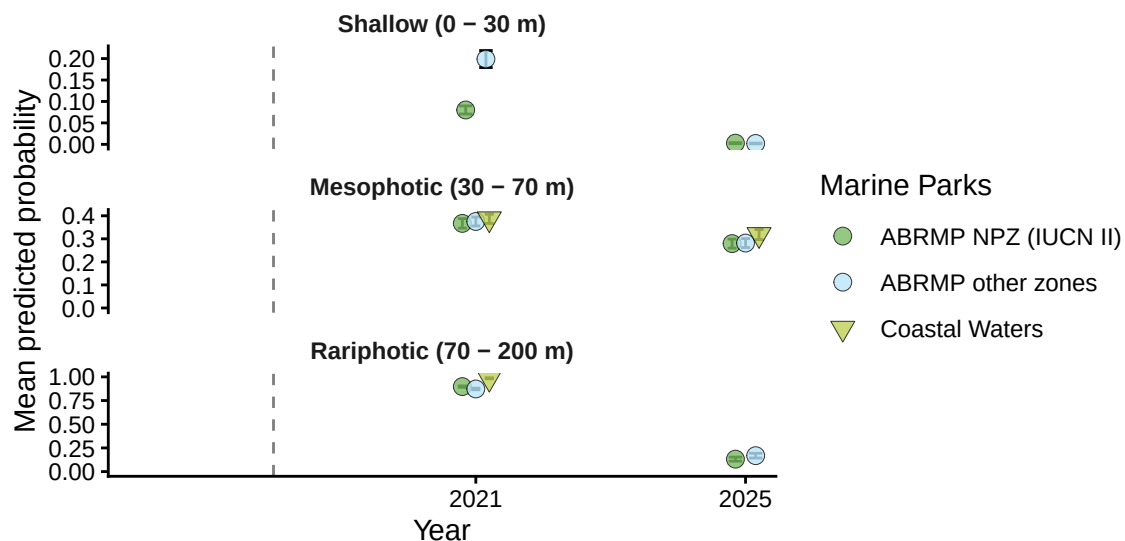


Figure 6: Sand: benchmark plots by zone and depth contour from stereo-BRUV and drop-camera groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

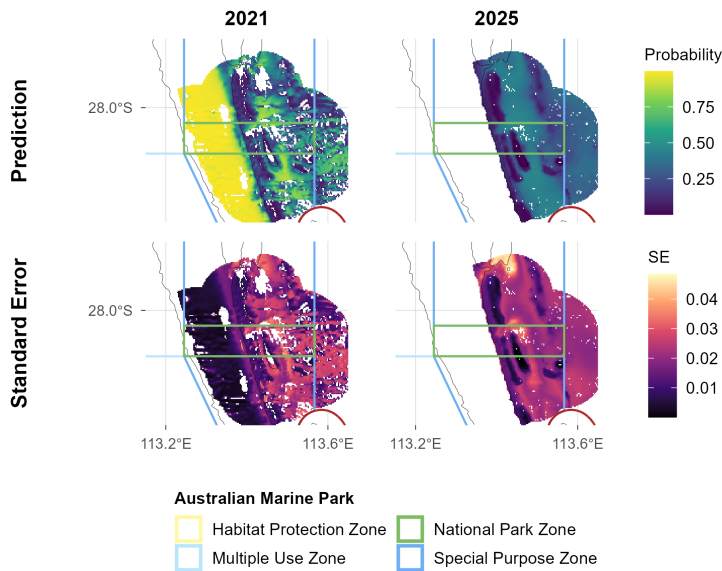


Figure 7: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Macroalgae

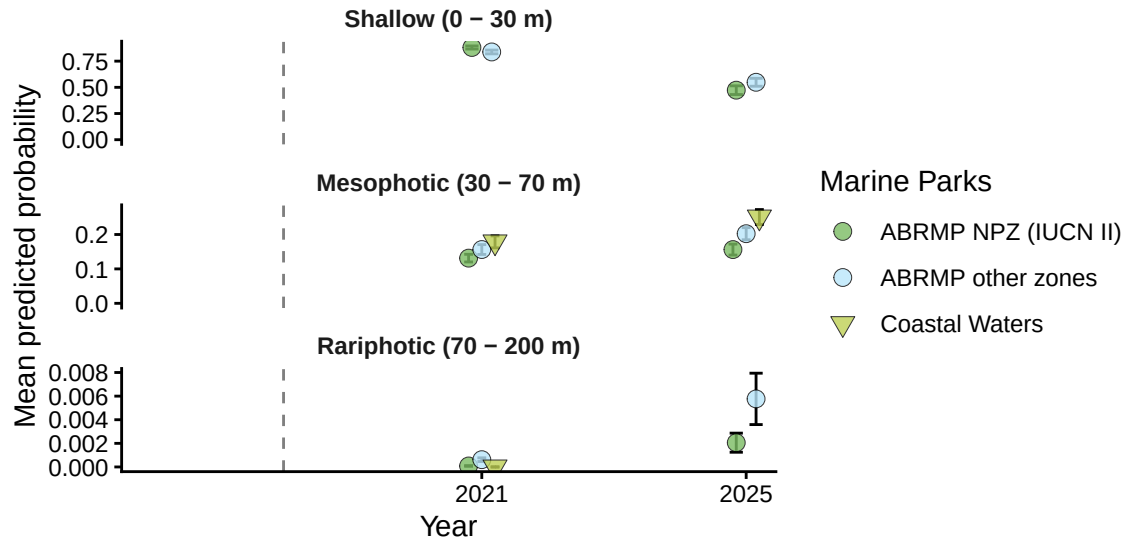


Figure 8: Macroalgae: benchmark plots by zone and depth contour from stereo-BRUV and drop-camera groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

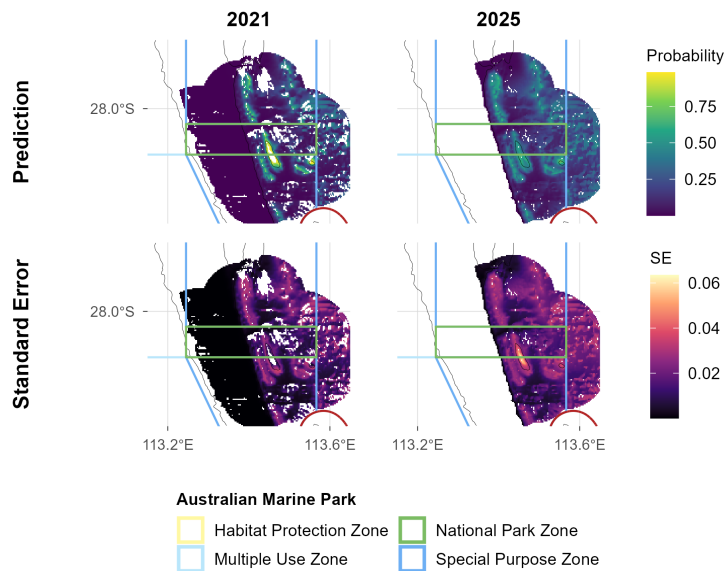


Figure 9: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Kelp

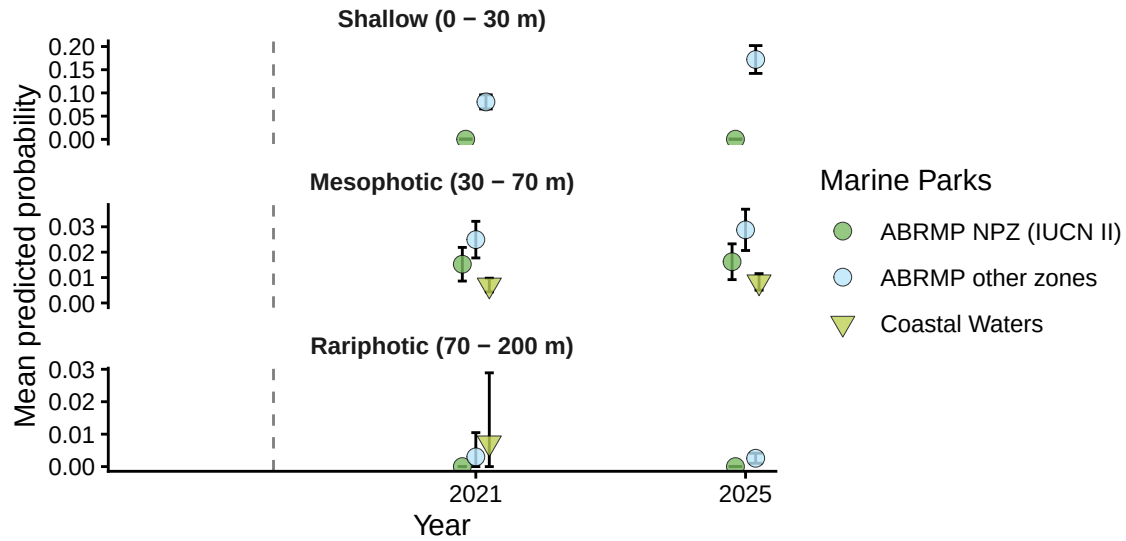


Figure 10: Kelp: benchmark plots by zone and depth contour from stereo-BRUV and drop-camera groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

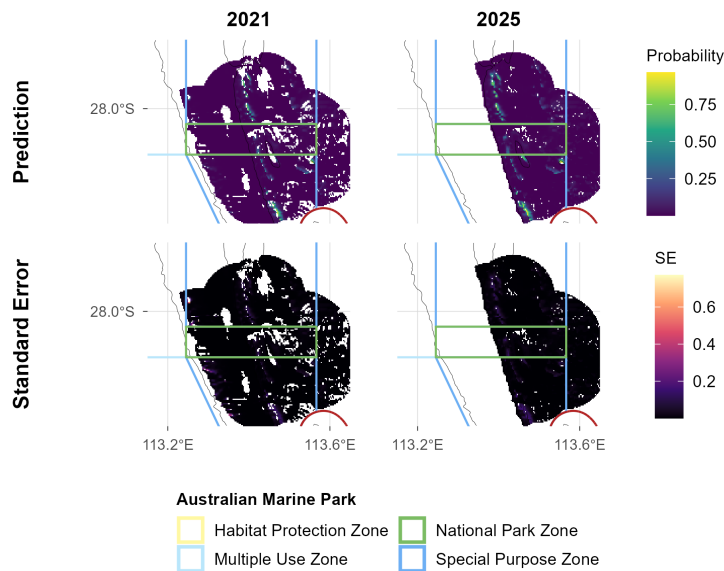


Figure 11: Kelp: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Rock

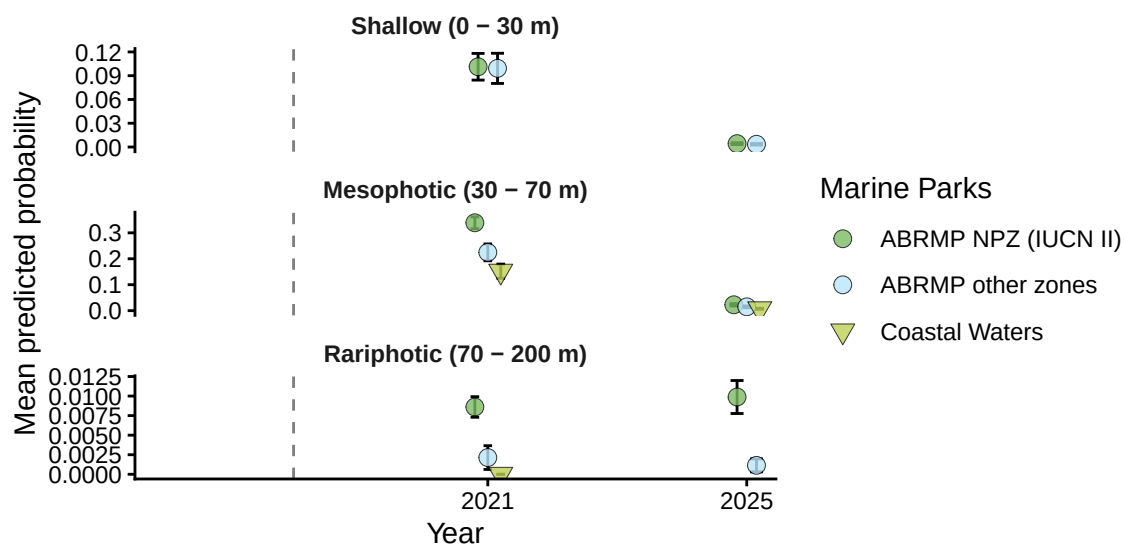


Figure 12: Rock: benchmark plots by zone and depth contour from stereo-BRUV and drop-camera groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

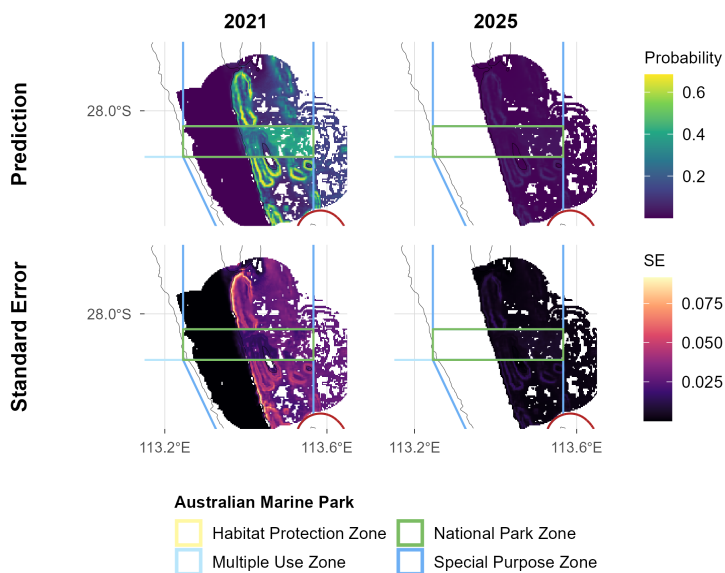


Figure 13: Rock: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Sessile invertebrates

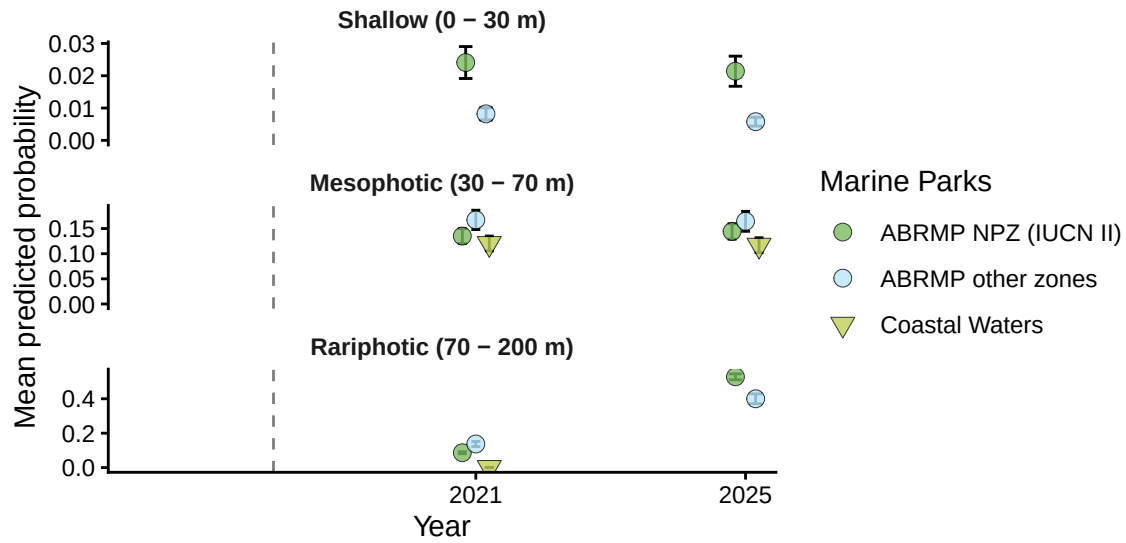


Figure 14: Sessile invertebrates: benchmark plots by zone and depth contour from stereo-BRUV and drop-camera groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

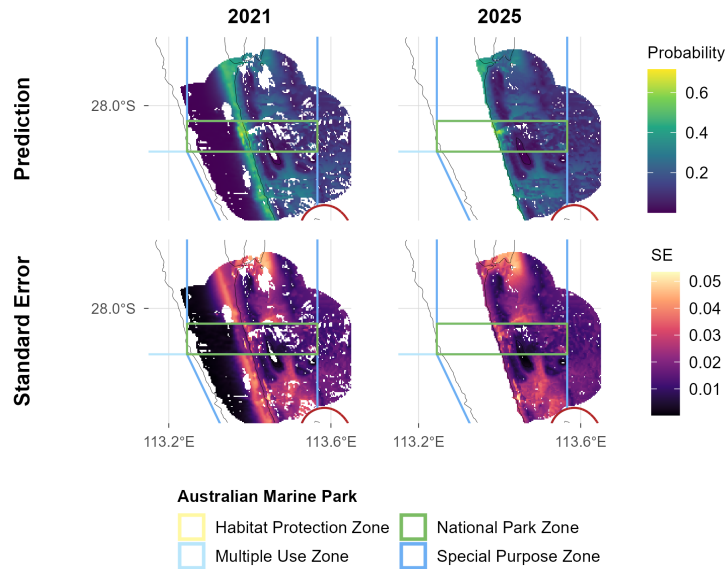


Figure 15: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Benchmarks of demersal fish natural values

### Survey effort

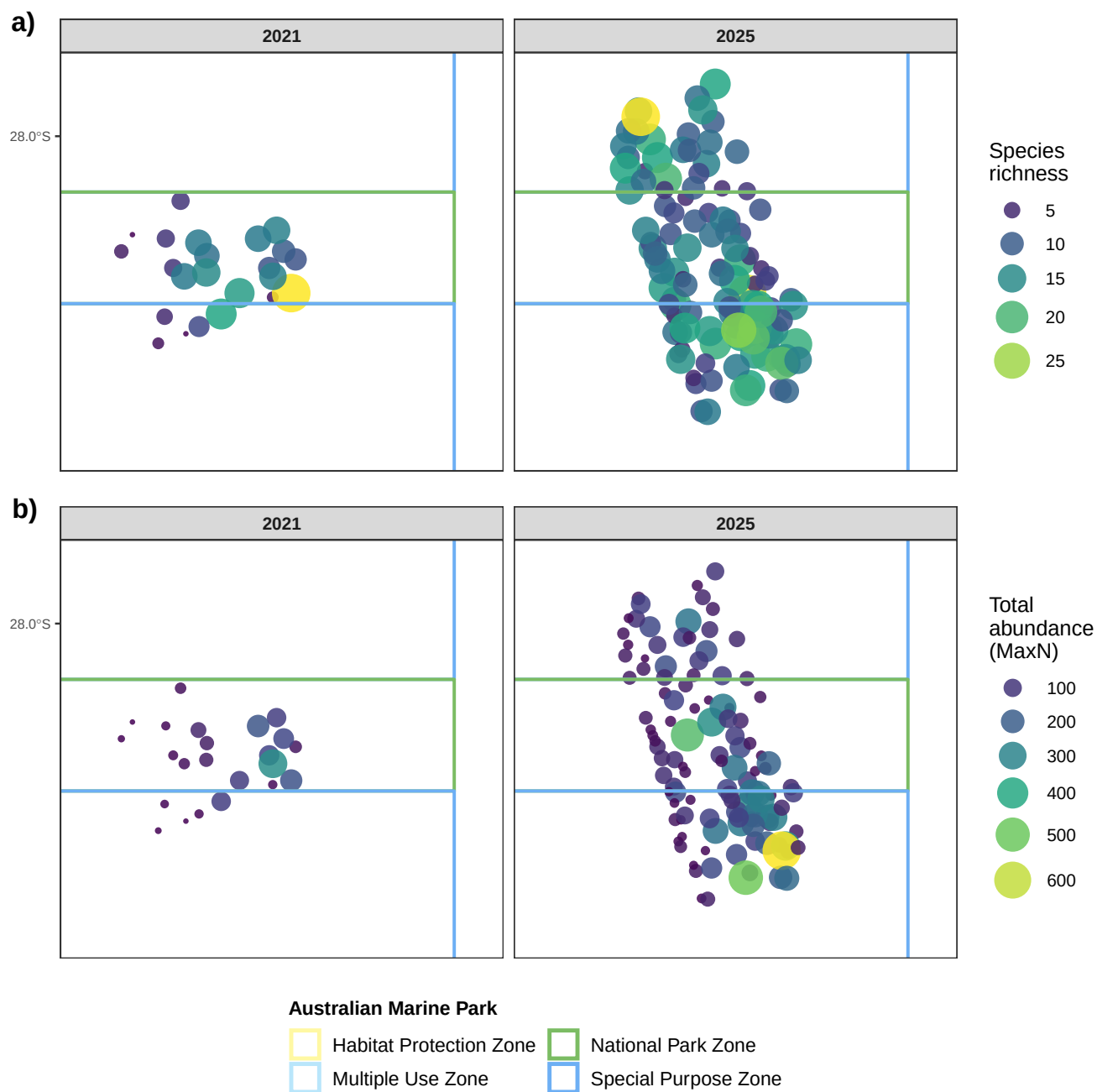


Figure 16: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV deployment, shown separately for the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

## Species richness

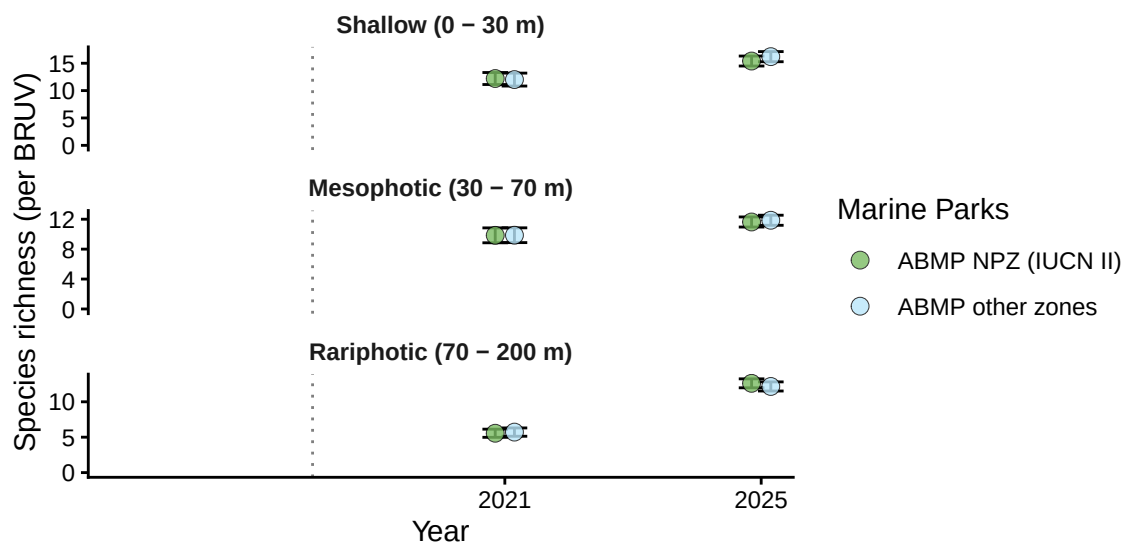


Figure 17.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

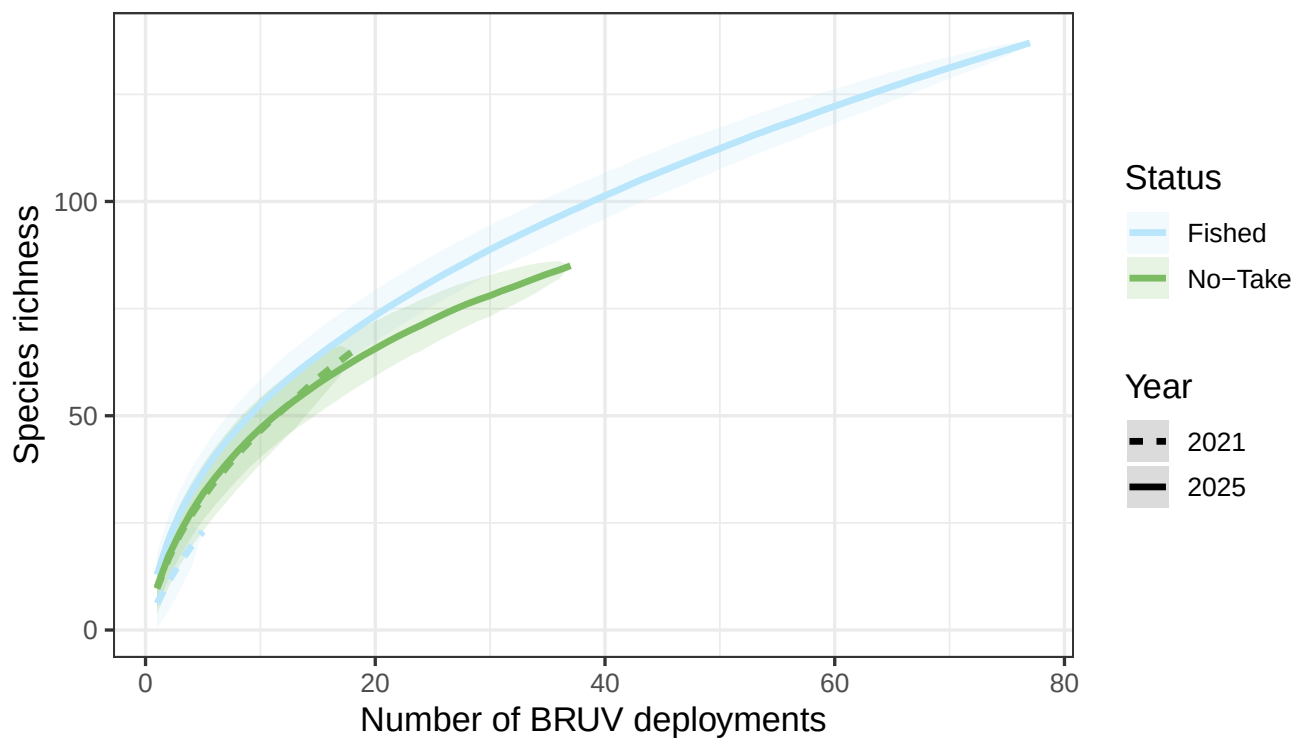


Figure 17.2: Demersal fish - Species richness: species accumulation curve by year and zoning status from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

## Species richness

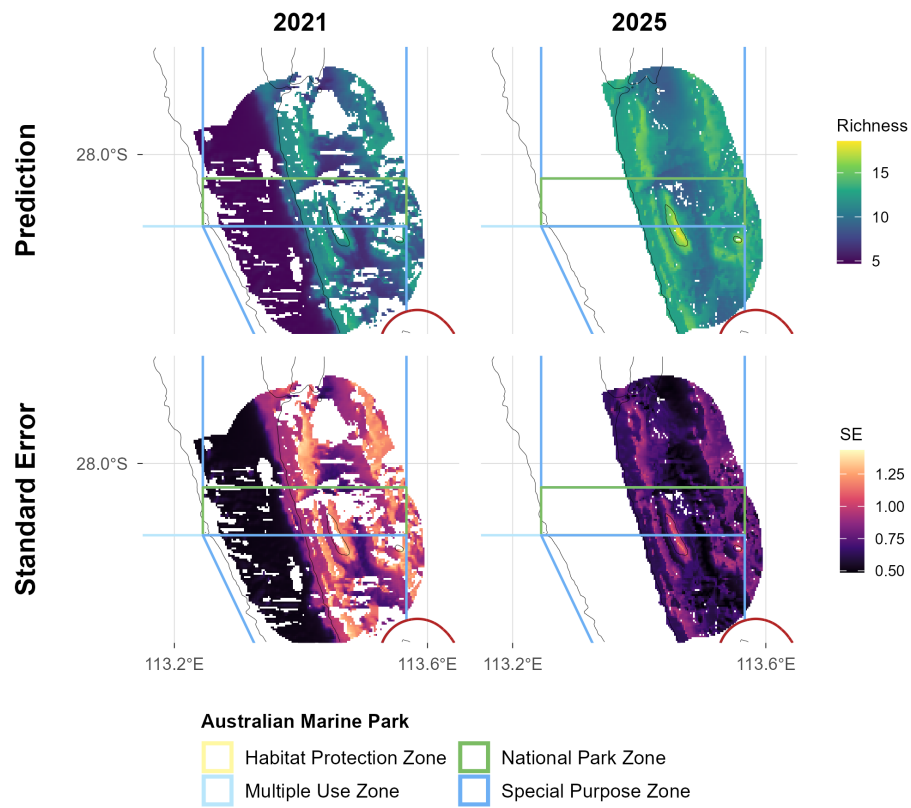


Figure 17.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing, shown separately for the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Total abundance

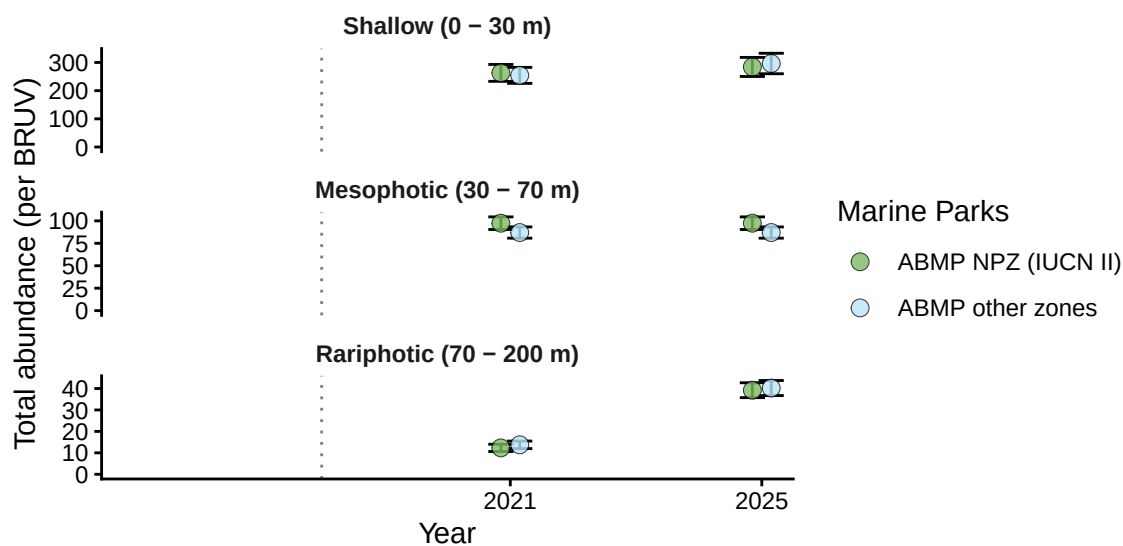


Figure 18.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

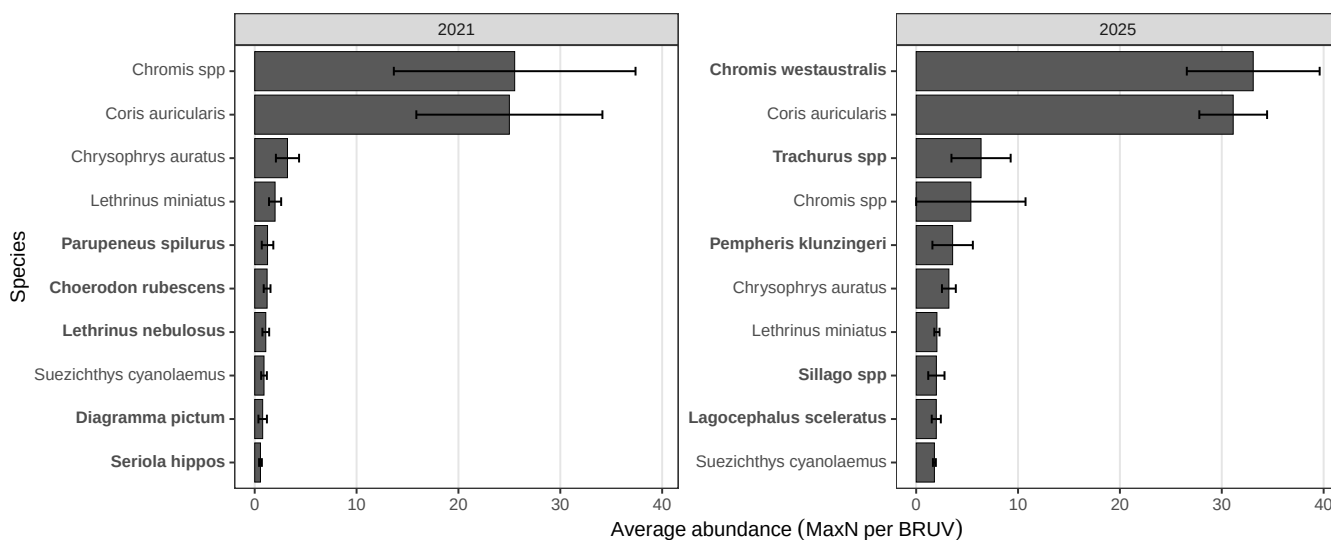


Figure 18.2: Demersal fish - Total abundance: top 10 most abundant species in each survey year from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). Species appearing in only one year's top 10 are shown in bold.

## Total abundance

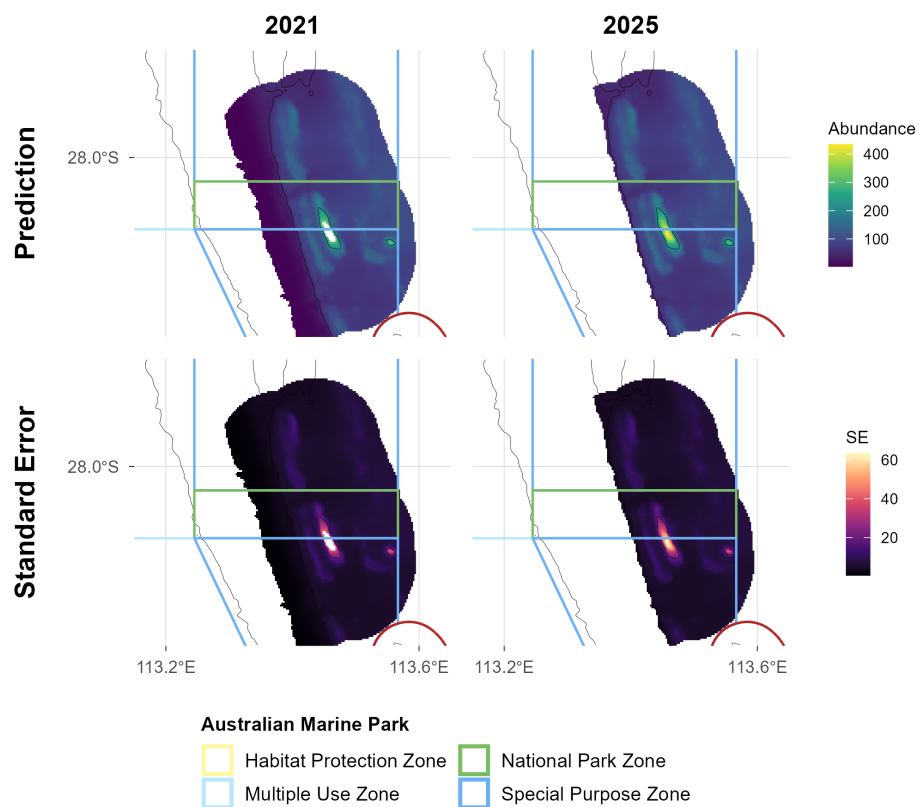


Figure 18.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing, shown separately for the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Large Reef Fish Index\*

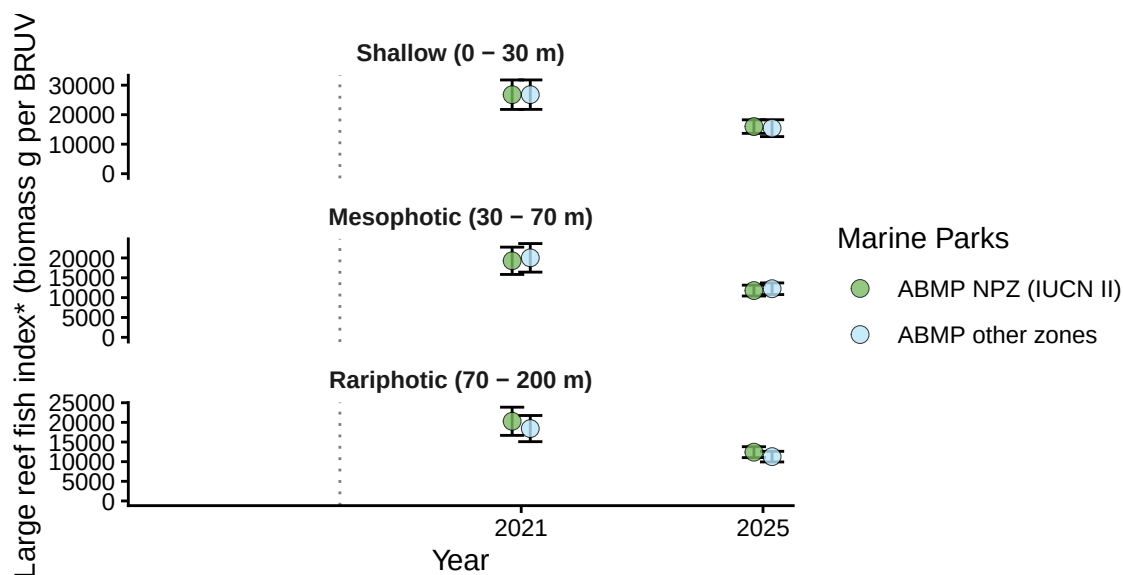


Figure 19.1: Demersal fish - Large Reef Fish Index\*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). \*Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

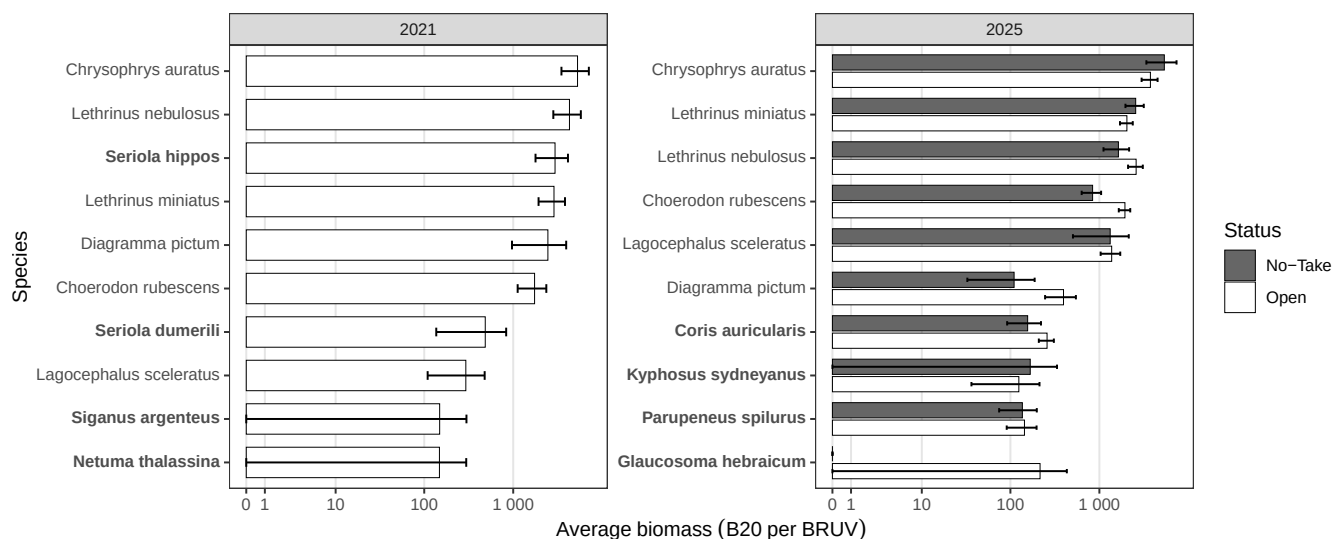


Figure 19.2: Demersal fish - Large Reef Fish Index\*: top 10 greatest biomass species in each survey year from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). The 2021 survey is shown pooled across zones because only five deployments were in fished waters.

## Large Reef Fish Index\*

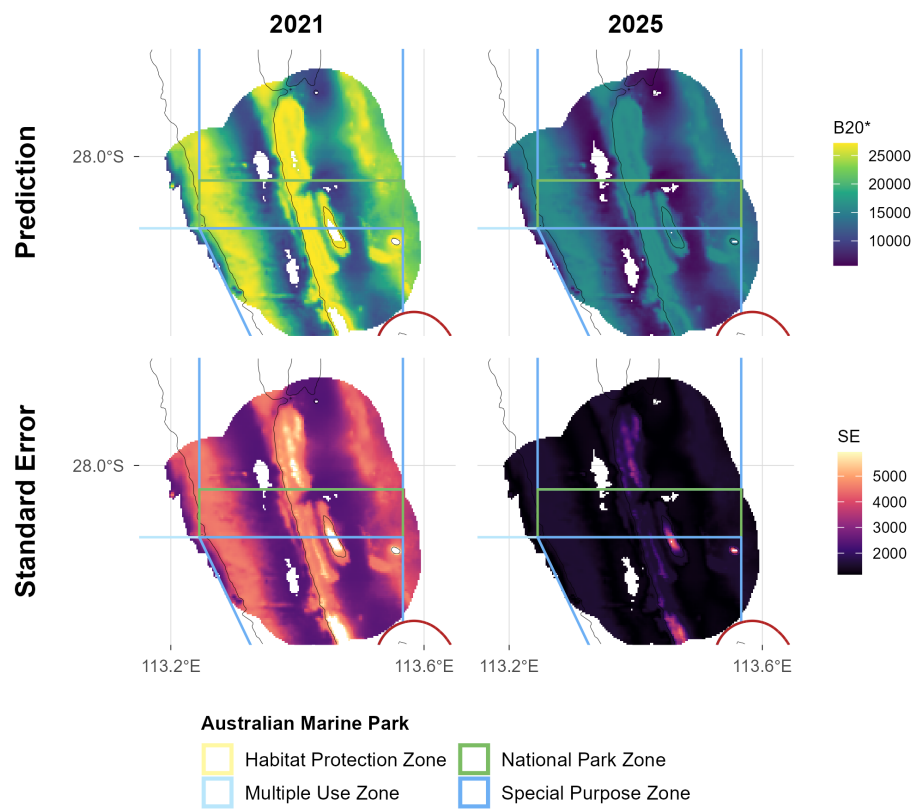


Figure 19.3: Demersal fish - Large Reef Fish Index\*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing, shown separately for the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

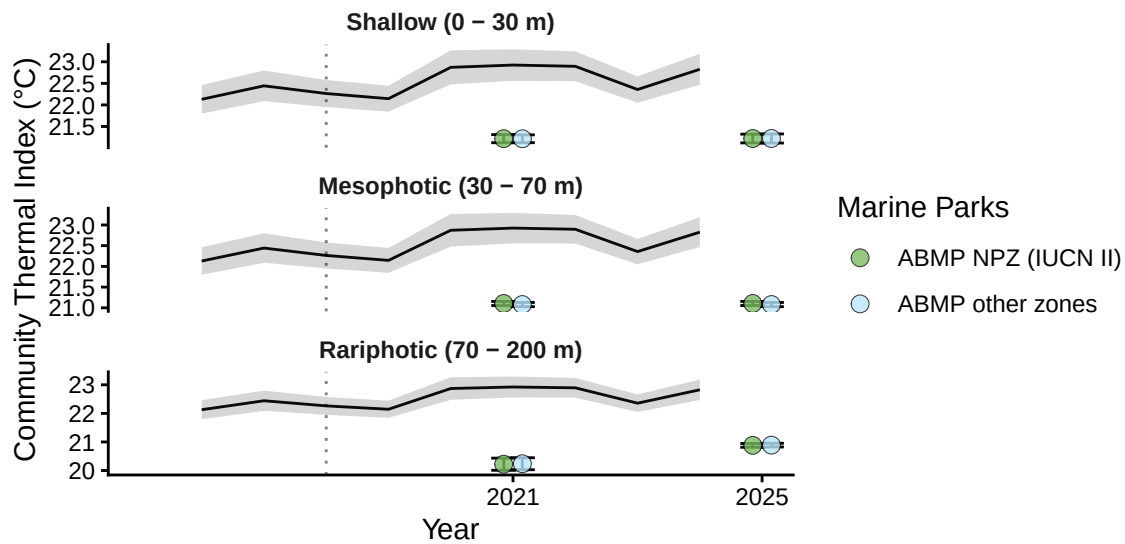


Figure 20.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with annual sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

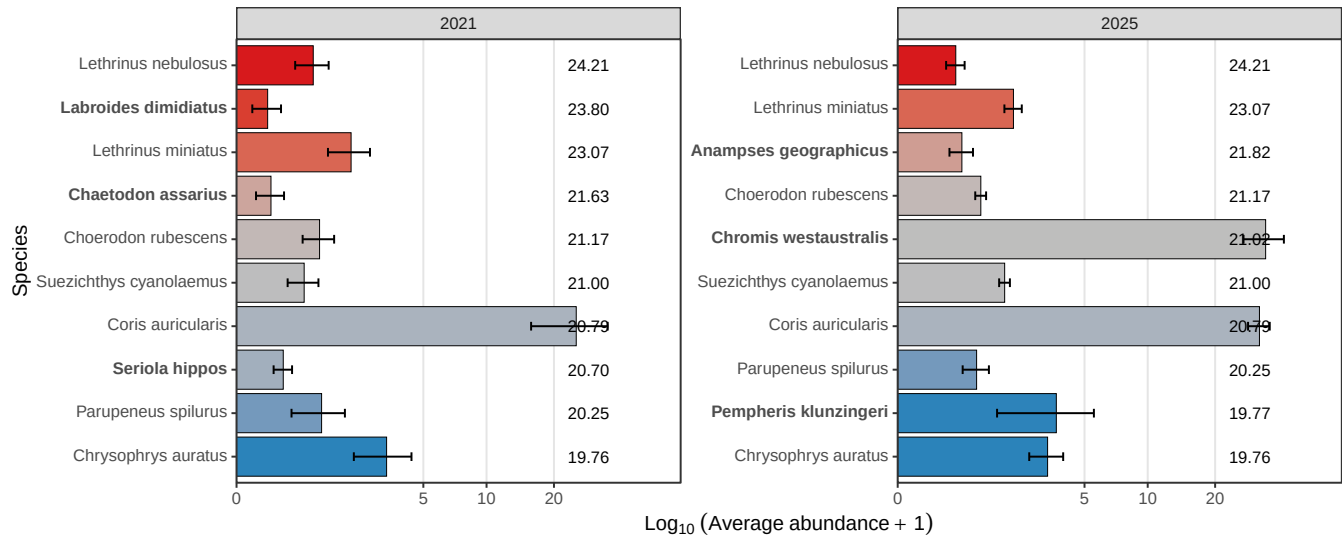


Figure 20.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of the top 10 most abundant species in each survey year from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

## Reef fish thermal index

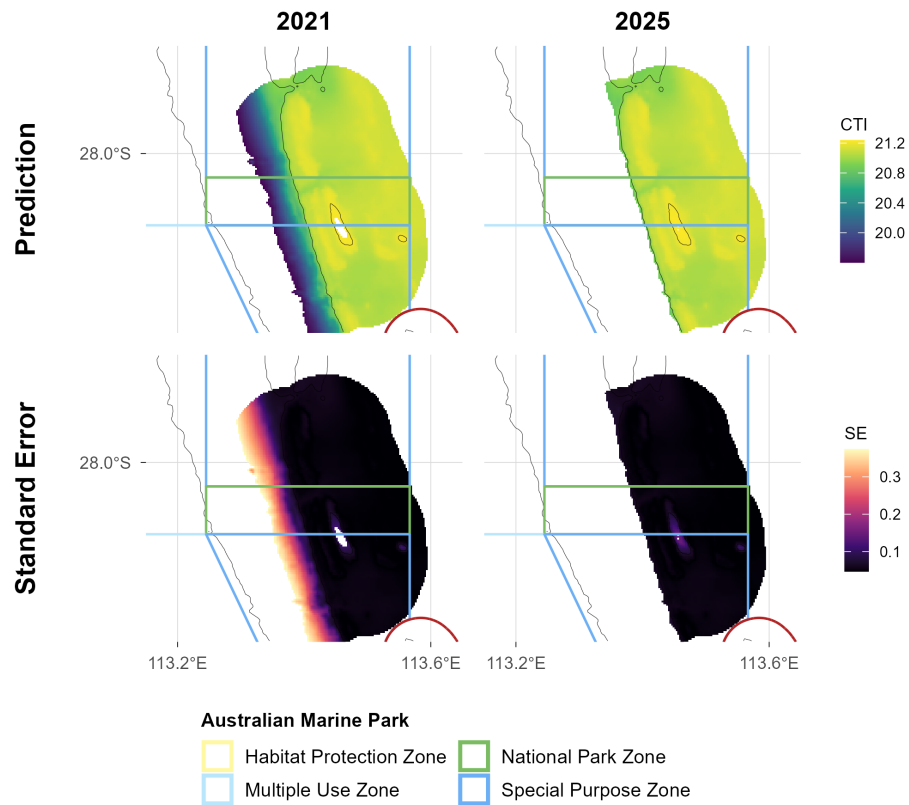


Figure 20.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing, shown separately for the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Threatened species

Table 2: Threatened species recorded in stereo-BRUV imagery across all survey years, showing IUCN Red List category, the number of samples each species was recorded in, and the total abundance (MaxN) summed across those samples. \*This species is also listed as ‘Conservation dependent’ on the EPBC Act List of Threatened Fauna. ^This species is also listed as ‘Vulnerable’ on the EPBC Act List of Threatened Fauna.

| Species                            | Common name             | Family         | IUCN category         | Samples | Total MaxN | Years recorded |
|------------------------------------|-------------------------|----------------|-----------------------|---------|------------|----------------|
| <i>Carcharhinus plumbeus</i>       | Sandbar Shark           | Carcharhinidae | Endangered            | 18      | 19         | 2021, 2025     |
| <i>Loxodon macrorhinus</i>         | Sliteye Shark           | Carcharhinidae | Near Threatened       | 5       | 6          | 2025           |
| <i>Sepia apama</i>                 | Giant cuttlefish        | Sepiidae       | Near Threatened       | 5       | 5          | 2025           |
| <i>Rhynchobatus australiae</i>     | Whitespotted Guitarfish | Rhinidae       | Critically Endangered | 4       | 4          | 2021, 2025     |
| <i>Taeniurops meyeri</i>           | Blotched Fantail Ray    | Dasyatidae     | Vulnerable            | 3       | 3          | 2021, 2025     |
| <i>Carcharhinus albimarginatus</i> | Silvertip Shark         | Carcharhinidae | Vulnerable            | 2       | 2          | 2025           |
| <i>Galeocerdo cuvier</i>           | Tiger Shark             | Carcharhinidae | Near Threatened       | 2       | 2          | 2021, 2025     |
| <i>Carcharhinus sorrah</i>         | Spot-tail Shark         | Carcharhinidae | Near Threatened       | 1       | 1          | 2025           |
| <i>Sphyrna lewini</i> *            | Scalloped Hammerhead    | Sphyrnidae     | Critically Endangered | 1       | 1          | 2021           |

## Threatened shark and ray species

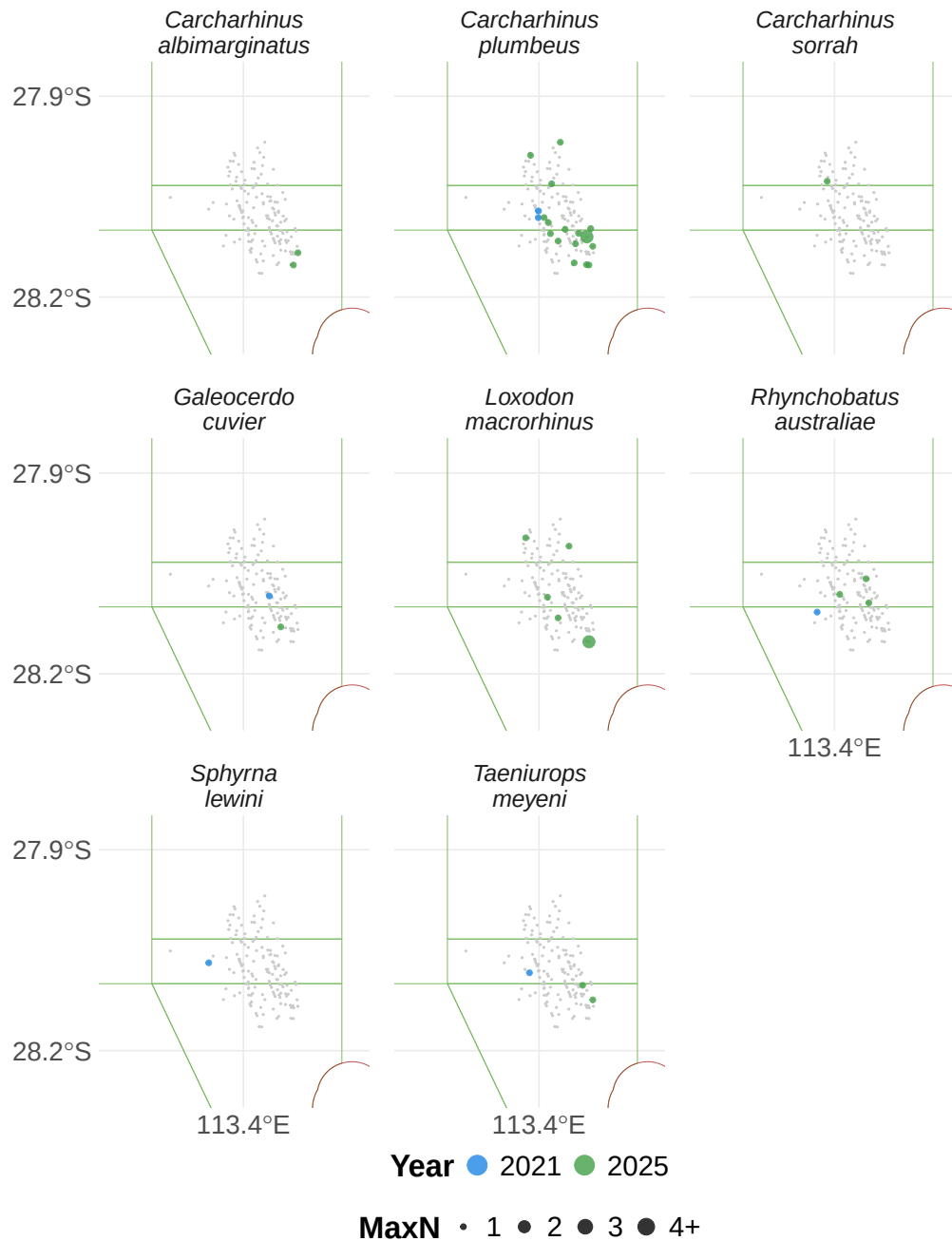


Figure 21.1: Occurrence of threatened, IUCN Red List category, sharks and rays in stereo-BRUV imagery, shown across all survey years. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

## Other shark and ray species

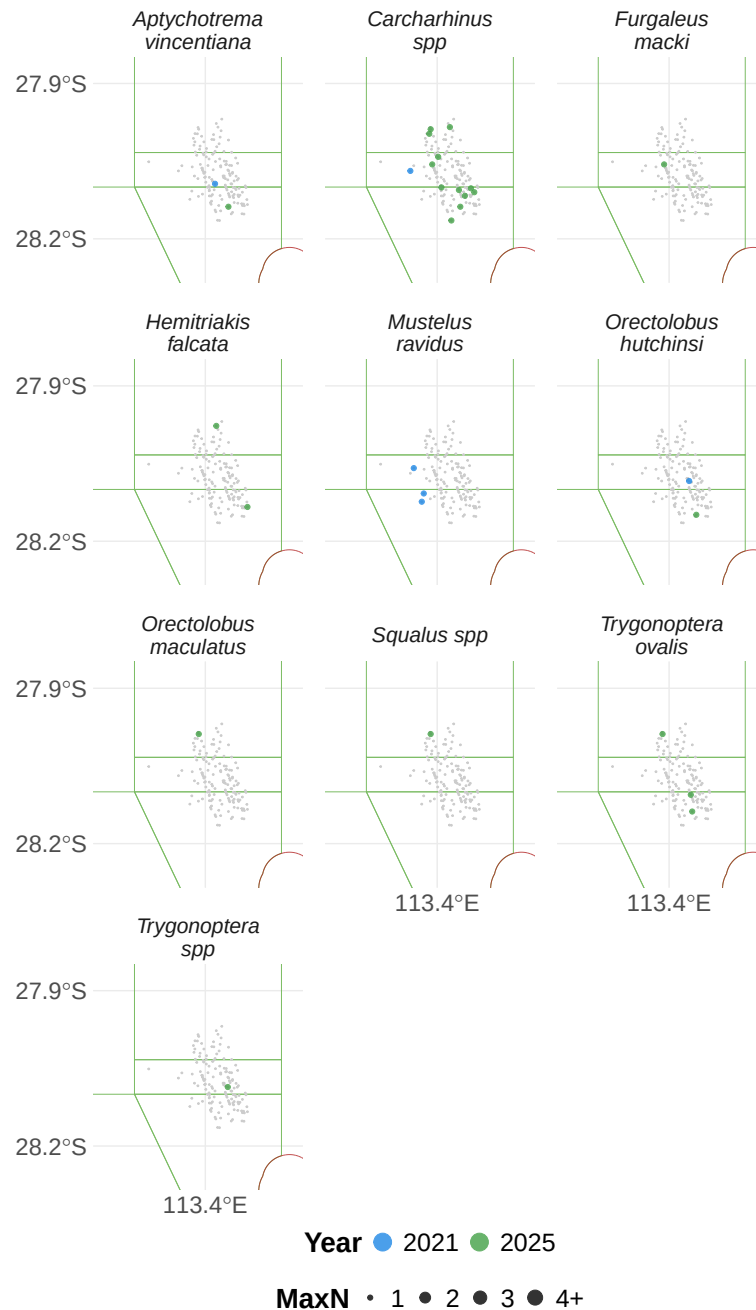


Figure 21.2: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery, shown across all survey years. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.